

## **THEORY AND MODELS IN ECOLOGY: A DIFFERENT PERSPECTIVE**

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### **ABSTRACT**

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Many widespread criticisms of ecological theory, especially theory relying on mathematical models, are based on misunderstandings the role of models in theory and of theory in the larger discipline of ecology. The naivete of some of these misunderstandings suggests that they are too seldom examined. In this paper, I address what I consider to be the most ill-conceived criticisms of ecological theory. I propose that the parallels between theoretical and empirical research are more profound than most people realize. Appreciation of these parallels will go a long way towards bridging the gap between empiricists and theoreticians in ecology.

### **INTRODUCTION**

“Ecology is awash with all manner of untested (and often untestable) models, most claiming to be heuristic, many simple elaborations of earlier untested models. Entire journals are devoted to such work, and are as remote from biological reality as are faith-healers.”

SIMBERLOFF, 1981

The use of mathematical models in theoretical ecology is seen by some ecologists today as a symptom of a malignancy infecting the entire discipline of ecology. I believe this diagnosis to be fundamentally mistaken. It arises from a misunderstanding of the nature of theoretical problems, from denial of their legitimacy as a part of ecology, from misapprehension of the place of mathematical models in the solution of theoretical problems, and from a surprising philosophical naivete about the limits of theory *and* empiricism in science. These confusions lead people to evaluate theoretical ecology on

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inappropriate grounds. In this note I propose a different perspective on theory and theorizing in ecology. To be as provocative as possible, I present my points as blatant assertions.

Lest there be confusion, let me preface my remarks with some disclaimers. As a Popperian falsificationist, I am not questioning the importance of experimental testing of theories, although I will take exception to a philosophy that views such testing as the only worthwhile thing to do with a theory. Nor am I trying to protect ecological theory from criticism, although I will argue that certain critiques are mistaken. There is quite as much bad theoretical research in ecology as there is bad empirical research; but probably not more. Nor am I trying to set 'theoreticians' against 'empiricists'. Quite the contrary: a better understanding of the nature of theoretical research should bring the two groups closer. In fact, I am going to continually point out parallels between theoretical and empirical research, parallels which, if appreciated, would make much of the criticism of theoretical ecology evaporate.

#### MODELS AND THEORY

*Models and theory are not equivalent. Theory can be developed without mathematical models, and models can be constructed for reasons having nothing to do with theory. In the long run, however, the incorporation of mathematical models usually improves the quality of ecological theory.*

Some very important ecological theory has been developed without using mathematical models at all. Some claim that the development of mathematical versions of these theories is a waste of time:

"The bad news is not so much that competition theory has failed to contribute to these discoveries as that ... much time has been wasted by ecologists as well as by theoreticians in pursuing this fruitless path."

SIMBERLOFF, 1982

I argue that, on the contrary, the incorporation of mathematics has generally contributed directly to our theoretical knowledge of ecological systems. Consider as an example the relation between the complexity and stability of ecosystems. The first statement of this theory was entirely verbal: Elton (1958) theorized (without using a single equation) that more complex ecosystems are more stable.

The first detailed mathematical investigation of the theory came 15 years later (May, 1973). When Pimm (1984, table 2) reviewed the subject a decade after that, he could tabulate over 30 theoretical studies on complexity and stability.

A look at the structure of Pimm's table, ignoring its contents, reveals something important about the role of models in ecological theory. Elton

had hypothesized that complex systems are more stable. But Pimm distinguishes two types of complexity (species richness and connectance), four types of stability (stability *sensu stricto*, resilience, persistence, and resistance) and three levels of resolution (individual species abundances, species composition, and total density). There are now 24 distinct aspects of the complexity–stability problem. How does species richness affect the resilience of species composition? How does connectance affect the persistence of individual species? None of these distinctions were, or could have been, drawn by Elton. Their importance became apparent only as the original verbal theory was studied using mathematical models.

This is an important contribution. It would remain so even if not a single empirical study had been carried out in the meantime. (No, I am not recommending that approach!) The theoretical studies demonstrated, if nothing else, that *the situation must be more complicated than Elton thought it was*. Neither complexity or stability is a simple matter, and the relations between the two must depend on how each is measured.<sup>1</sup> Haldane's (1964) eloquent defense of theoretical population genetics makes a similar argument.

This is an example of *theoretical knowledge*. Though these models say nothing about the stability or the complexity of any individual ecosystem, they tell us something very important indeed about the *possible* relations between stability and complexity. It is by this exploration of possible patterns that theory makes one of its major contributions to ecology, by suggesting new ways of looking at problems:

"No practical biologist interested in sexual reproduction would be led to work out the detailed consequences experienced by organisms having three or more sexes; yet what else should he do if he wishes to understand why the sexes are, in fact, always two? The ordinary mathematical procedure in dealing with any actual problem is, after abstracting what are believed to be the essential elements of the problem, to consider it as one of a system of possibilities infinitely wider than the actual, the essential relations of which may be apprehended by generalized reasoning, and subsumed in general formulae, which may be applied at will to any particular case considered."

FISHER, 1958

"There is again a wide misunderstanding of the function of theoretical studies in population biology. It is *not* the function of theory to describe what has happened in a particular instance. Only observation can do that. The purpose of theoretical studies in population biology is *to set limits*, to say, 'This can happen; this cannot. This process will be extremely slow or of extremely small magnitude as compared to that process.' Theoretical population biology is the science of the *possible*; only direct observation can yield a knowledge of the actual."

LEWONTIN, 1968

<sup>1</sup> Of course, this is not the only contribution of this body of theory; see Pimm (1984)

Theoretical knowledge, like empirical knowledge, is fallible. It is *not* guaranteed to be correct. I have heard it argued that theory should not be considered ‘part of ecology’ until it has been ‘validated’. This is both philosophically and historically naive; no part of any science, empirical or theoretical, is ever validated. All our knowledge is conjectural. Everyone recognizes that theories are refuted and replaced, but the history of any significant problem in any discipline reveals that experimental results are also contradicted by later experiments, and have their interpretations drastically changed. To exclude fallible theories from ecology while admitting fallible experiments is absurd. But, understanding how theory fits into the larger context of ecology requires an appreciation of the problems that theoretical models attempt to solve.

## THEORETICAL PROBLEMS

*Theoretical problems have a legitimate, objective existence in ecology.*

Perhaps the greatest obstacle to understanding the role of theory is the failure to recognize that theoretical studies attempt to solve *theoretical problems*, and that these problems are legitimate part of ecology.

Theoretical problems are those arising from a body of theory (or sometimes from the lack of one). One important theoretical problem is: ‘It this theory really true in nature?’ (Are more complex ecosystems really more stable?) This problem cannot be solved by theory alone; it requires experimental or observational tests of the predictions of the theory, and the answer is always fallible. However, other, equally important theoretical problems arise any time a theory begins to develop. Some of these problems ask questions about the theory itself; they cannot be answered by empirical investigation.

As an example, consider food web structure. Analyses of published food web data (Cohen, 1978; Briand, 1983; Briand and Cohen, 1984; etc.) reveal statistical patterns in the proportions of different trophic levels, the scaling relationships of trophic links, the distribution of food chain lengths, etc. Most of these observations fit within no current theoretical framework. The first theoretical problem is, then, simply to find a theory which includes the observed patterns among its predictions. What is the simplest such theory? How simple can such a theory be a still make new predictions?

Cohen and Newman (1985) set out to construct and compare several *simple* models for food web construction. The simplest generates food webs by specifying only the number of species and the density of trophic links. It then assigns trophic links among the species at random. It fails to predict the observed patterns.

Their most successful model (the ‘cascade model’) is only marginally more complex, but does an astonishing job of duplicating both the qualitative and quantitative patterns of observed webs. In the cascade model, the species are ordered; each species can be potentially fed on by all species higher in the order, but none lower. Trophic links are assigned at random among these potential links, using the observed link density. Real food webs are certainly not assembled this way – nor do Cohen and Newman claim that they are. But the cascade model is a solution to an important theoretical problem: what is the simplest model which can predict the observed phenomenon? It is thus already a contribution to *ecological* knowledge. It provides a starting point for both new kinds of observations and new theories. The latter will almost certainly render it obsolete – at which point it will have made an even greater contribution.

The question of equilibrium provides another example. By the early 1970’s ‘predator-mediated coexistence’ (situations in which two or more competing prey species coexist only in the presence of a predator) was well-documented. Attempts had been made to build models, based on Lotka–Volterra theory, to predict this phenomenon. The analyses sought conditions under which two competing prey were unstable, while two prey and a predator were stable. It appeared that this was possible, but required a more delicate balancing of parameters than seemed likely in nature (May, 1973).

This raised a theoretical problem: could predation mediate coexistence without stabilizing the equilibrium? The answer (Caswell, 1978) is yes; demonstrating it required a change in the theoretical criteria for coexistence, a distinction between local and regional coexistence, and a simple simulation model. The results showed that the phenomenon could be generated in a robust fashion, if the predator disturbed the equilibrium of its prey within a patch, and the prey were able to disperse between patches.

This study made several theoretical contributions to ecological knowledge, none of which hinged on its ‘validation’. It demonstrated the *possibility* of one important outcome of a class interactions in a particular kind of environment, it suggested ways of looking at empirical data on predator-mediated coexistence (e.g. the importance of the spatial scale of experiments), and it suggested further theoretical problems.

In each of these examples a problem was raised by the then-current state of ecological theory. Far from being “as remote from biology as faith-healing” (Simberloff, 1981), such theoretical problems have an objective <sup>2</sup> existence, and are legitimate part of the discipline of ecology.

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<sup>2</sup> Those who are fond of citing Karl Popper in these sorts of arguments should study his discussions (e.g. Popper, 1972) of the objective reality of what he terms “World 3”, which contains, inter alia, theories, criticisms of theories, and theoretical problems.

The following is a beginning of a list of types of theoretical problems:

(1) Exploring the consequences of a theory. To determine what is possible from a theoretical model can be a contribution to ecological knowledge, regardless of the eventual decision about the truth or falsity of the model in any specific case. This activity generates lots of papers exploring the dynamic consequences of particular equations. Rather than protesting that those papers do not contain empirical tests, it should be understood that most of them are addressing very different questions.

(2) Demonstrating the connection between apparently unrelated theories. For example, the demonstration that the bifurcation properties of several different discrete density-dependent models are generic to any smooth one-humped map of the interval (May, 1976) was a contribution to ecological knowledge, quite independent of the question of which (if any) discrete model gives the best fit to a stock–recruitment relationship.

(3) Evaluation of the robustness, or lack thereof, of theoretical results, by examining different models for the same phenomenon.

(4) Developing mathematical methods for the analysis of theoretical models.

(5) Finding the *simplest* model capable of generating an observed pattern.

(6) Designing empirical and/or experimental tests of the theory. Everybody admits that this is an important problem. As I will discuss in the next section, too many think it is the only important problem.

Of course, not every paper that analyzes ecological equations is, or is intended to be, a contribution to ecological theory. Many examine the equations for mathematical, not ecological, reasons. These papers should be (and usually are) published in mathematical, not ecological journals, and they should be judged on mathematical, not ecological grounds.

The standards of criticism for theoretical ecological research are not well developed. As long as ecologists continue to deny the legitimacy of theoretical problems, it will remain inherently impossible to decide whether any specific paper makes a significant contribution to one.

#### METHODOLOGICAL MISTAKES AND SLOPPY PHILOSOPHY

*We have been lulled into several serious methodological errors. These errors appear with disturbing frequency in reviews of theoretical manuscripts and research proposals, are invoked as gospel in discussion of theoretical work, and are taught to students as if they were a prescription for how to do ecology right. They are not.*

The three most important of these errors are:

(1) The *only* important thing to do with theories is to test them.

“What must we do to make ecology healthy? First, insist that mathematical or verbal theory without direct, rigorous field testing no longer be recognized as a significant contribution... Heuristic value is unacceptable as a justification for publishing.”

SIMBERLOFF, 1981

Yes, rigorous testing of theory is important. This prescription, however, is unreasonable and its implementation would be a disaster. To see why, consider that everyone admits that science progresses by the integration of theory and experiment. I know not a single theoretical ecologist who dreams of a data-free world, nor a single empiricist whose secret fantasy is the mindless accumulation of raw facts unencumbered by theory. Given the interpenetration of theory and experiment, why not turn Simberloff's prescription on its head, and insist that field or experimental data, unaccompanied by the development of significant new theoretical models, no longer be recognized as a significant contribution?

Strictly interpreted, this would bring to a grinding halt the publication of most experimental data in ecology. An empiricist can, of course, say, ‘I test theories; I'm quite happy to leave their development to others who are more qualified, or who enjoy it more.’ I personally accept this attitude, but those who hold it must allow theoreticians the corresponding liberty to work on the development of theory, leaving experimental tests to others, perhaps more qualified, or who enjoy it more.

An alternative argument is, ‘While the development of new theory is indeed a worthy goal, it is unreasonable to expect *every* piece of experimentation to be accompanied by significant new theory. It usually requires a series of experiments, each informed by its predecessors, perhaps conducted by different people working in different systems, before significant new theories will be inspired.’ This argument applies equally well to theoretical research. The development of rigorous field testing usually requires a series of models, each informed by its predecessors, developed by different people within different mathematical frameworks. To demand that each step in the process be accompanied by rigorous field testing would bring a halt to the development of theory as surely as the demand that each experiment be accompanied by significant new theory would halt experimental research.

(2) The refute a theory is (or ought to be) to abandon it.

Failure to recognize that theoretical models attempt to solve theoretical problems automatically makes it impossible to decide if any particular use is appropriate. The only position left is to claim that a refuted model should never be used.

"Despite continued indications that certain models (e.g. logistic, Lotka-Volterra, stock-recruitment) are not suitable for most situations, many theoreticians and managers continue to use them."

HALL and DEANGELIS, 1985

Karl Popper is often (incorrectly) cited in support of this position. Popper himself makes no such mistake. He draws an explicit distinction between recognizing that a theory has been refuted by failing an experimental test and deciding to abandon work on it.

"One of his [Lakatos's] more confusing moves is his repeated contention that I conflate the refutation of a theory with its rejection; that is, with the necessity for people to abandon the theory, and more especially to abandon working on it. Indeed, while the former is, given the acceptance of a refuting state of affairs, a matter of logic, the latter is a question of methodology, and will depend among other things, on what alternative theories are available. ... That I do not conflate even admitted falsity with the need to abandon a theory may be seen from the fact that I have frequently pointed out that Einstein regarded general relativity as false, yet as a better approximation to the truth than Newton's gravitational theory. He certainly did not 'abandon' it."

POPPER, 1974

(3) Assumptions are evil, and anything not explicitly included in a model is implicitly assumed to be unimportant in nature.

It is popular to criticize theoretical models for 'making too many unrealistic assumptions', and in particular for failing to include factors known to be important. Now, it is certainly legitimate to ask: 'Does this model include factor *X*, which is known to be important in certain situations?' Far too often, however, the model is written off as irrelevant if the answer is negative, even though the inclusion of *X* may be completely inappropriate for the question the model is addressing! The failure to include a factor in a theoretical model is a legitimate criticism only if it can be argued that its inclusion would materially affect the answer to the question under consideration.

Consider the analogous situation in empirical studies. Wondering if my empirical colleagues might know something I did not, I collected the data shown in Fig. 1, at an extremely empirical poster session on Terrestrial Ecology, Pine Forests, and Nutrient Cycling at the IVth International Congress of Ecology in Syracuse, NY, August 1986. For a haphazard sample of these posters, I tabulated the *number of factors* included in the design of the experiments reported.<sup>3</sup> *Two* factors? *Three* factors? Surely, no pine

<sup>3</sup> In fact, this tabulation is generous in that I counted time as a factor. Most of the three-factor studies were manipulations of two factors (e.g., light and nutrients) with a response followed over time. A more conservative approach would view the time series itself as the response; such experiments would then become two-factor studies.



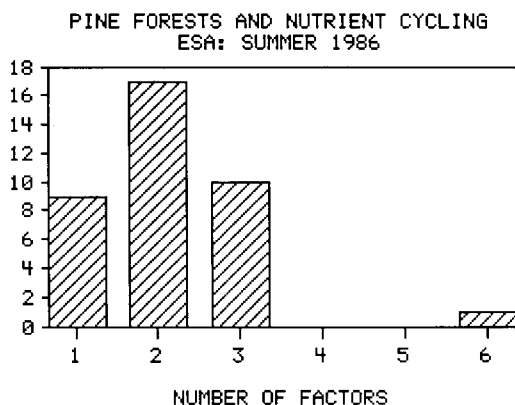


Fig. 1. Frequency distribution of the number of experimental factors included in a haphazard sample of a poster session on Terrestrial Ecology, Pine Forests, and Nutrient Cycling at the IVth International Congress of Ecology, Syracuse, NY, August 1986.

forest ecologist would argue that nutrient cycling is completely determined by two or three factors, and that all others are irrelevant. Nor would any pine forest ecologist be likely to agree that by reporting an experiment including only two or three factors, he or she was assuming that no other factors were important. In fact, the absence of other factors in the experimental design would be accepted as a criticism of such a study only if it could be argued that those factors might change the answers to the questions under investigation.

The situation with theoretical models is exactly parallel. Just as the number of factors in an experiment may be limited by the inability to manipulate large number of experimental units, so the incorporation of too many factors into a model may render it mathematically intractable. Just as large factorial experiments can bury an investigator in high-order interactions that are impossible to interpret and which obscure the main effects of interest, so huge simulation models can obscure the action of any of the factors included.

Even more important, however, any experimentalist recognizes that no experiment is the last word on the subject; that eventually it will be understood in the context of other experiments which manipulate other factors. The same is true for theoretical models. Any experiment, and any model, must leave some factors out of consideration (even factors *known* to be important). This fact of life is used as a critique of experimentation much less frequently than as a critique of theory, but such criticisms are not unknown:

"I believe that in the United States the progress of marine biology was retarded for 50 years by the introduction of experimental methods into biology. Experiments ... attempt to relate behavior quantitatively to one or more variables in the situation, all other

variables supposedly being kept under control. In this sense, experiments are always *unnatural*, and subject to the question: how adequate are the results when applied to situations in nature?"

REDFIELD, 1958

## A DIFFERENT PERSPECTIVE

I want to propose a perspective on models and theory which clarifies the role of models in theory and of theoretical problems in ecology as a whole. It also helps eliminate the methodological errors listed above. I propose an analogy: *models are to theoretical problems as experiments are to empirical problems.*

- Models are constructed to improve our *understanding* of a theoretical problem, just as experiments are conducted to improve our understanding of an empirical problem. There are many routes to understanding (quantitative prediction is only one; others include providing new perspectives on old problems, suggesting new observations, manipulation of ‘what-if’ scenarios, and suggesting new problems), and correspondingly many kind of models and experiments. The ultimate goal, however, is always the same.
- The factors included in a model are *not* necessarily the only ones important, just as the factors manipulated in an experiment are not necessarily the only ones important.
- Neither the model nor the experimental system need duplicate every detail of the real world.
- Models and experiments are both parts of larger contexts; no single study will, or should be expected to, settle an issue completely.
- Critical discussion of models should focus on the danger of “artifacts” introduced by the structure of the model, just as critical discussion of an experiment focuses on artifacts introduced by the experimental design.
- Experimental artifacts are avoided by conducting more and different experiments. Modelling artifacts are to be avoided analogously.

Anyone who has listened to experimental biologists argue about the interpretation of their results or the adequacy of their methods will recognize that this proposal will not discourage criticism. However, stressing the importance of the problem situation (theoretical or experimental, as the case may be) and the parallels between the models and experiments will lead to a better understanding and more appropriate criticism of ecological theory.

For example, Simberloff (1982) has written an interesting critique of competition theory. Among other things, he contrasts the “meager accomplishments” of competition theory with the “considerable progress” made by detailed experimental studies (“investigations”), rich in natural historical

detail, of specific communities. He criticizes competition theory because:

"My guess is that for virtually every interspecific competitive effect that has been predicted by a theoretician, the opposite effect has been predicted, at least in principle, by some other theoretician (or even the same theoretician)."

SIMBERLOFF, 1982, p. 242

But later, discussing the accomplishments of the investigation approach, he says (and not as a criticism!):

"First, it is clear that integral to all of these systems are biological idiosyncrasies, ... An understanding of each system was thus accessible only to those who studied the natural history very intensively, and results from each system are not easily extrapolated to other systems."

SIMBERLOFF, 1982, p. 250

In other words, one might say that for virtually every interspecific competitive effect revealed by empirical investigation, the opposite effect has been revealed by some other investigators, or even the same investigator. What was a failing of theoretical studies is apparently not when it appears as a result of empirical investigation. Whether this multiplicity of effects is a sign of our ignorance (i.e., if we only knew what to look for, we would find a simple and consistent pattern of competitive effects) or represents a real state of affairs is perhaps an open question. However, the appearance of diverse, idiosyncratic, and apparently contradictory results must be an indictment of both theory and empiricism, or of neither.

## IN CONCLUSION

I have argued in this paper for the legitimacy of theoretical problems, and for a new perspective on the use of models as a tool for investigating such problems. What is needed is a more pluralistic and tolerant understanding of the bridge between theory and experiment. As theory is developed and explored, it will suggest new perspectives, new interpretations, new observations, new theoretical and empirical problems. All of these are contributions to ecology, regardless of whether they are accompanied by the ability to make accurate quantitative predictions of specific situations<sup>4</sup>. By understanding the objectives of theory, empiricists will be better able to take advantage of these developments. By clarifying the theoretical purposes of

<sup>4</sup> Let me clarify a point which has been misunderstood by some readers of drafts of this paper. There is nothing wrong with quantitative prediction. It is indeed something to strive for. In fact, it is becoming clear that there interesting theoretical problems involve with the attempt to incorporate more mechanistic detail into theoretical models. Such developments are to be encouraged. What I am arguing against is a particularly myopic view which maintain the successful prediction (or 'validation') in the sine qua non of ecological theory.

their work, theoreticians can help its eventual integration with empirical work.

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#### REFERENCES

- Briand, F., 1983. Environmental control of food web structure. *Ecology*, 64: 253–263.
- Briand, F. and Cohen, J.E., 1984. Community food webs have scale-invariant structure. *Nature London*, 307: 264–266.
- Caswell, H., 1978. Predator-mediated coexistence: a nonequilibrium model. *Am. Nat.*, 112: 127–154.
- Cohen, J.E., 1978. Food Webs and Niche Space. Monographs in Population Biology, 2. Princeton University Press, Princeton, NJ, 189 pp.
- Cohen, J.E. and Newman, C.M., 1985. A stochastic theory of community food webs. I. Models and aggregated data. *Proc. R. Soc. London B*, 224: 421–448.
- Elton, C.S., 1958. The Ecology of Invasions by Animals and Plants. Methuen, London, 181 pp.
- Fisher, R.A., 1958. The Genetical Theory of Natural Selection (2nd Edition). Dover, New York, 291 pp.
- Haldane, J.B.S., 1964. A defence of beanbag genetics. *Perspect. Biol. Med.*, 1: 343–359.
- Hall, C.A.S. and DeAngelis, D.L., 1985. Models in ecology: paradigms found or paradigms lost? *Bull. Ecol. Soc. Am.*, 66: 339–346.
- Lewontin, R.C., 1968. Population Biology and Evolution. Syracuse University Press, Syracuse, NY, 205 pp.
- May, R.M., 1973. Stability and Complexity of Model Ecosystems. Monographs in Population Biology, 6. Princeton University Press, Princeton, NJ, 235 pp.
- May, R.M., 1976. Simple mathematical models with very complicated dynamics. *Nature*, 261: 459–467.
- Pimm, S.L., 1974. The complexity and stability of ecosystems. *Nature London*, 307: 321–326.
- Popper, K.R., 1972. Objective knowledge: an evolutionary approach. Oxford University Press, Oxford, 380 pp.
- Popper, K.R., 1974. Replies to my critics. In: P.A. Schilpp (Editor), The Philosophy of Karl Popper. Open Court Press, LaSalle, IL, pp. 961–1200.
- Redfield, A.C. 1958. The inadequacy of experiment in marine biology. In: A. Buzzati-Traverso (Editor), Perspectives in Marine Ecology. University of California Press, Berkely, CA, pp. 17–26.
- Simberloff, D., 1981. The sick science of ecology: symptoms, diagnosis, and prescription. *Eidema*, 1: 49–54.
- Simberloff, D., 1982. The status of competition theory in ecology. *Ann. Zool. Fenn.*, 19: 241–253.